



COM

736

Database Systems and Data Analytics (Week 10)

Identifiers and Identifier Systems

Data Identification

First step in data management

Least understood Big Data issue

Distinguishes the data object

Links the information

Reflects the organisational model

Add value

Improper identification will result in failure

Data Identification

De-identification

Removes the **links**

Personal information,
demographics etc.

Re-identification

Re-establish the links

Associate to the
individual.

Data Identifier System

Object identifier

- Alphanumeric string
- Associated with a data object

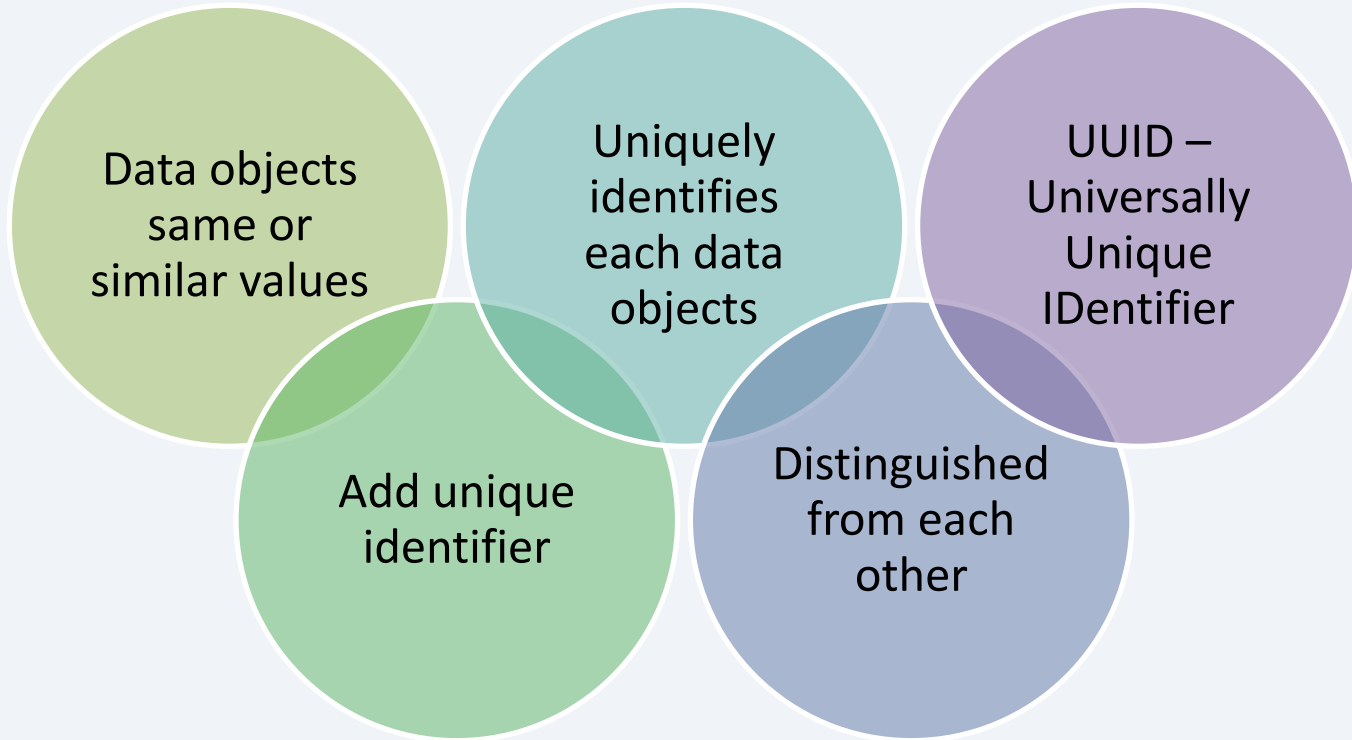
Identifier System

- Environment where identifiers stored and used

Properties

- Completeness, Uniqueness, Authenticity, Security and many more

Unique Identifier



Bad Identifiers

Names

Unlikely to be unique

Not recommended

Name may change

Embedded information

Can extract information

Collect confidential information

Your student ID

Driving licence number

De-identification

Removing individuality

Anonymise the data

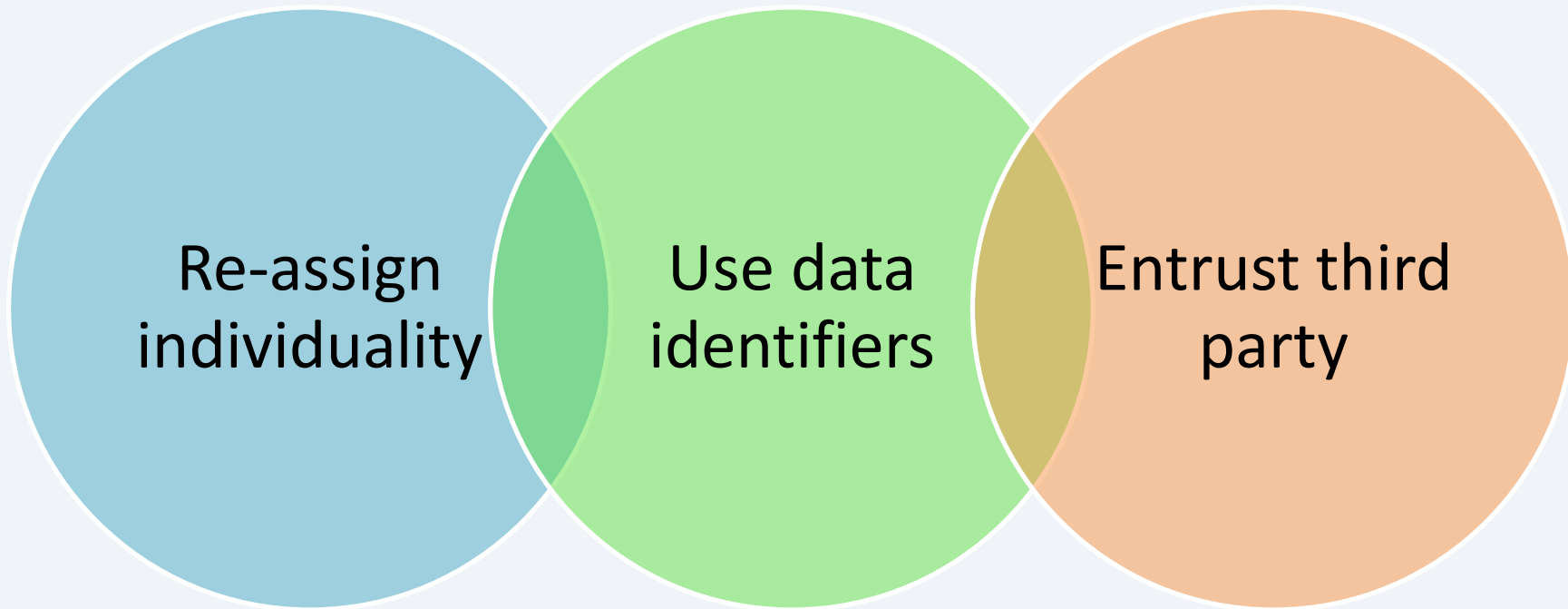
Keep the unique identifier

Protect confidentiality

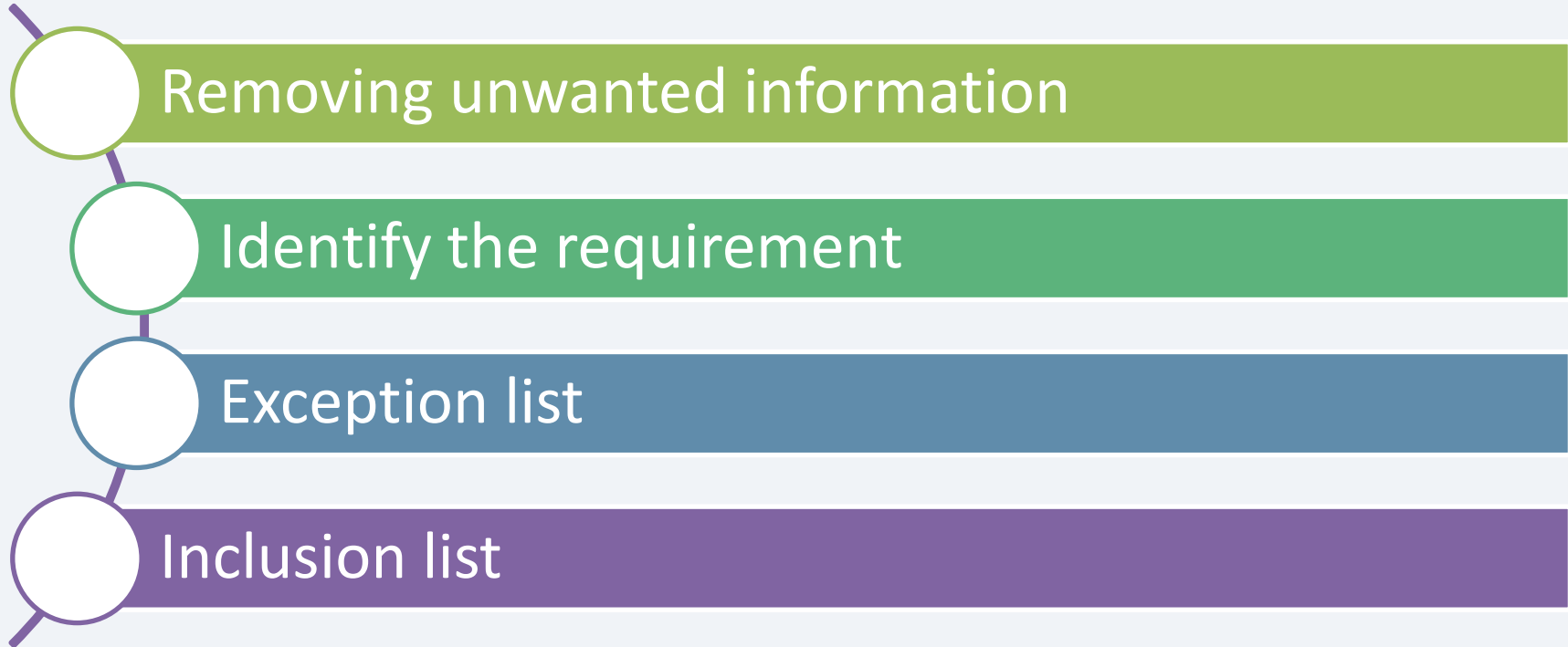
Avoid biased output

De-identify properly identified objects

Re-identification



Data Scrubbing



Next

Metadata, Classification and Ontologies

A decorative graphic consisting of a horizontal bar with a blue border, divided into segments of blue, teal, green, yellow, and orange, ending in a large orange arrow pointing to the right.

Data about the data